(ii) Types of Hyperbola with centre at (h, k)

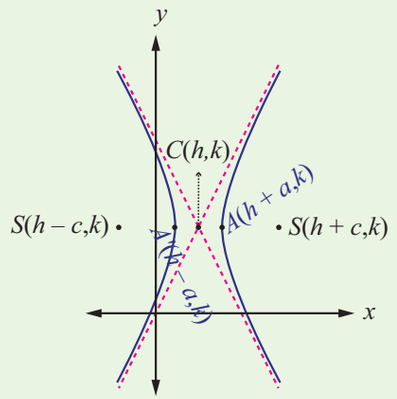


Fig. 5.29
(a) transverse axis parallel to the x-axis

(a) Transverse axis parallel to the x-axis.

The equation of a hyperbola with centre C (h,k) and transverse axis parallel to the x-axis (Fig. 5.29) is given by $\frac{(x-h)^2}{a^2}-\frac{(y-k)^2}{b^2}=1$.

The coordinates of the vertices are $A(h+a,k)$ and $A'(h-a,k)$. The coordinates of the foci are $S(h+c,k)$ and $S'(h-c,k)$ where $c^2=a^2+b^2$.

The equations of directrices are $x=h\pm\frac{a}{e}$.

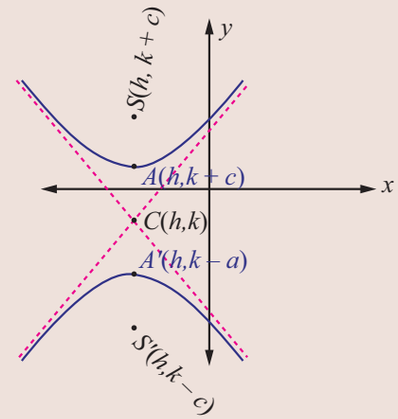


Fig. 5.30
(b) transverse axis parallel to the y-axis

(b) Transverse axis parallel to the y-axis

The equation of a hyperbola with centre $C(h,k)$ and transverse axis parallel to the y-axis (Fig. 5.0) is given by

$$\frac{(y-k)^2}{a^2}-\frac{(x-h)^2}{b^2}=1.$$

The coordinates of the vertices are $A(h,k+a)$ and $A'(h,k-a)$.The coordinates of the foci are $S(h,k+c)$ and $S'(h,k-c)$, where $c^2=a^2+b^2$.

The equations of directrices are $y=k\pm\frac{a}{e}$.

Remark

(1) The circle described on the transverse axis of hyperbola as its diameter is called the auxiliary circle of the hyperbola. Its equation is $x^2+y^2=a^2$.

(2) The absolute difference of the focal distances of any point on the hyperbola is constant and is equal to length of transverse axis. That is, $|PS-PS'|=2a$. (can be proved similar that of ellipse)

So far we have discussed four standard types of parabolas, two types of ellipses and two types of hyperbolas. There are plenty of parabolas, ellipses and hyperbolas whose equations cannot be classified under the standard types, For instance consider the following parabola, ellipse, and hyperbola.

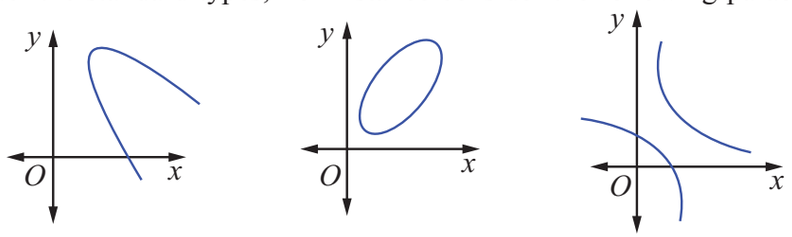


Fig. 5.31

By a suitable transformation of coordinate axes they can be represented by standard equations.

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